Continuous-Time Markov Chains (CTMCs)

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References

- S.M. Ross, “Stochastic Processes”: Chapter 5
- R. Wolff: Chapter 4
- S.M. Ross, “Introduction to Probability Models”: Chapter 6
- A. Kumar, “Discrete Event Stochastic Processes”: Chapter 4

Definition

- Consider a continuous-time process $\{X(t), t \geq 0\}$, which takes on values in the set of non-negative integers
- The process $\{X(t), t \geq 0\}$ is said to be a CTMC if for all $s, t \geq 0$, and non-negative integers $i, j, x(u), 0 \leq u \leq s$:
  1) $P\{X(t+s) = j | X(s) = i, X(u) = x(u), 0 \leq u < s\} = P\{X(t+s) = j | X(s) = i\}$
- Property 1) is the Markovian property
- In addition, if $P\{X(t+s) = j | X(s) = i\}$ is independent of s, then the CTMC is said to have *homogeneous* transition probabilities
- Throughout, we only study CTMCs with homogeneous transition probabilities

Distribution of Amount of Time Spent in a State

- Suppose:
  - a CTMC enters state i at some time, say time 0,
  - the process does not leave state i until time s
- Probability that process remains in state i during the interval $[s, s+t]$:
  - equals unconditional probability that it stays in state i for at least t time units
  - by Markovian property
- That is, if τ_i denotes the amount of time that the process stays in state i before making a transition into a different state, then:
  - $P\{\tau_i > s + t | \tau_i > s\} = P\{\tau_i > t\}, \forall s, t \geq 0$
- So, the r.v. τ_i is memoryless and hence must be exponentially distributed

Properties of a CTMC

- From above discussion, it follows that a CTMC is a stochastic process with the properties that each time it enters state i:
  a) the amount of time it spends in that state before making a transition into a different state is exponentially distributed with rate, say, v_i, and
  b) when the process leaves state i, it will next enter state j with some probability, say P_{ij}, where:
     - $\sum_{j \neq i} P_{ij} = 1$
- If $v_i = \infty$, then state i called an *instantaneous* state since when entered, it is instantaneously left
- If $v_i = 0$, then state i called an *absorbing* state since once entered, it is never left
- Throughout, we assume that $0 \leq v_i < \infty$

Properties of a CTMC (contd.)

- Recall:
  - τ_i denotes the amount of time that the process stays in state i before making a transition into a different state
- τ_i and the next state visited must be independent r.v.s
- Reason:
  - Suppose next state visited were dependent on τ_i
  - Then information as to how long the process has already been in state i would be relevant to prediction of next state
  - This would contradict Markovian property
- Relation between a CTMC and a semi-Markov process:
  - CTMC is a semi-Markov process with $F_{ij}(t) = 1 - e^{-v_i t}$

Regular CTMCs

- A CTMC is said to be *regular* if, w.p. 1, the number of transitions in any finite length of time is finite
- E.g. of a nonregular CTMC:
  - $P_{i,i+1} = 1$ and $v_i = i^2$
  - With positive probability, this CTMC makes an infinite number of transitions in a large enough time interval of length $t > 0$
    - **Proof**:
      - By Markov's inequality, $P(\sum_{i=1}^{\infty} \tau_i \geq t) \leq \frac{1}{t} E(\sum_{i=1}^{\infty} \tau_i) = \frac{1}{t} \sum_{i=1}^{\infty} \frac{1}{i^2} < 1$, for $t > \sum_{i=1}^{\infty} \frac{1}{i^2}$
- Henceforth, we assume that all CTMCs considered are regular

Transition Rates

- Let:
  - $q_{ij} = v_i P_{ij}$, for all $i \neq j$
- *Interpretation*: Since v_i is rate at which process leaves state i when it is in state i, and P_{ij} is probability that it then goes to state j, it follows that:
  - q_{ij} is the rate when in state i that the process makes a transition to state j
- Hence, q_{ij} is called *transition rate* from i to j
- Note that:

  1) $P_{ij} = \frac{q_{ij}}{v_i} = \frac{q_{ij}}{\sum_{k \neq i} q_{ik}}$
- Suppose the CTMC is presently in state i
- For $j \neq i$, let T_{ij} be the time until a transition from i to j would occur if next state were j
- Then it can be shown that:
  - for each $j \neq i$, T_{ij} is exponentially distributed with rate q_{ij}, and
  - T_{ij}, $j \neq i$, are independent r.v.s
- Equation 1) is consistent with fact that if $X_1, \ldots, X_n$ are independent and exponential r.v.s with rates $\mu_1, \ldots, \mu_n$, respectively, then $P\left(X_j = \min(X_1, \ldots, X_n)\right) = \frac{\mu_j}{\mu_1 + \cdots + \mu_n}$
- T_{ij} can be thought of as "*timers*" for different states $j \neq i$; CTMC transitions from state i to the state j whose timer goes off before timer of any other state $k \neq i$
- Let:
  - $P_{ij}(t) = P\{X(t+s) = j | X(s) = i\}$

Example 1

- Consider a shoe shine shop with two chairs, 1 and 2
- Customer upon arrival goes to chair 1, where his/ her shoes are cleaned and polish is applied
- After this is done, customer moves on to chair 2, where the polish is buffed
- Service times at the two chairs are independent r.v.s that are exponentially distributed with rates μ_1 and μ_2, respectively
- Potential customers:
  - arrive in accordance with Posson process with rate λ,
  - enter the shop only if both chairs are empty
- This process is a CTMC with following states:
  - 0: system is empty
  - 1: customer in chair 1
  - 2: customer in chair 2
- Transition rates:
  - $v_0 = \lambda$, $v_1 = \mu_1$, and $v_2 = \mu_2$
- Transition probabilities:
  - $P_{01} = P_{12} = P_{20} = 1$

Example 2

- Consider a Poisson process $\{N(t), t \geq 0\}$ with rate λ
- This process is a CTMC with states:
  - $0,1,2, \ldots$
- Transition rates:
  - $v_i = \lambda$, for all i
- Transition probabilities:
  - $P_{i,i+1} = 1$, for all i

Example 3

- Consider an M/M/1 queue with arrival rate λ and service rate μ
- It is a CTMC with states:
  - $0,1,2,\ldots$
- Transition rates:
  - $v_i = \begin{cases} \lambda, & \text{if } i = 0, \\ \lambda + \mu, & \text{if } i \geq 1. \end{cases}$
- Transition probabilities:
  - $P_{0,1} = 1$
  - $P_{i,i+1} = \frac{\lambda}{\lambda+\mu}$ and $P_{i,i-1} = \frac{\mu}{\lambda+\mu}$, for all $i \geq 1$

Birth and Death Processes

- A birth and death process is a CTMC with states $0,1,2,\ldots$ for which transitions from state i can only go to either state $i-1$ or state $i+1$
- Hence:
  - $q_{ij} = 0$ whenever $|i-j| > 1$
- State of process can be thought of as representing size of some population:
  - when state increases by 1, we say that a birth occurs,
  - when state decreases by 1, we say that a death occurs
- Let:
  - $\lambda_i = q_{i,i+1}$ for $i \geq 0$: called birth rates
  - $\mu_i = q_{i,i-1}$ for $i \geq 1$: called death rates
- Transition rates:
  - $v_i = \begin{cases} \lambda_i + \mu_i, & \text{for } i \geq 1, \\ \lambda_0, & \text{for } i = 0. \end{cases}$
- Transition probabilities:
  - $P_{0,1} = 1$ and $P_{i,i+1} = \frac{\lambda_i}{\lambda_i + \mu_i} = 1 - P_{i,i-1}$ for $i \geq 1$

Example

- Consider an M/M/s queue, where s is a positive integer, with:
  - arrival rate: λ,
  - service rate of each server: μ
- It is a birth and death process with states:
  - $0,1,2, \ldots$
- Birth rates:
  - $\lambda_n = \lambda$, for $n \geq 0$
- Death rates:
  - $\mu_n = \begin{cases} n\mu, & \text{if } 1 \leq n \leq s, \\ s\mu, & \text{if } n > s. \end{cases}$
- Transition rates:
  - $v_n = \begin{cases} \lambda, & \text{if } n = 0, \\ \lambda + n\mu, & \text{if } 1 \leq n \leq s, \\ \lambda + s\mu, & \text{if } n > s. \end{cases}$
- Transition probabilities:
  - $P_{0,1} = 1$
  - $P_{n,n+1} = \frac{\lambda}{\lambda+n\mu}$ and $P_{n,n-1} = \frac{n\mu}{\lambda+n\mu}$, for all $1 \leq n \leq s$,
  - $P_{n,n+1} = \frac{\lambda}{\lambda+s\mu}$ and $P_{n,n-1} = \frac{s\mu}{\lambda+s\mu}$, for all $n > s$

Pure Birth Processes

- A birth and death process is said to be a *pure birth* process if:
  - $\mu_n = 0$ for all n
- E.g.: Poisson process:
  - birth rate $\lambda_n = \lambda$ for all $n \geq 0$
- Another example:
  - Yule process

Example: Yule Process

- Consider a population in which:
  - each member acts independently and gives birth at rate λ,
  - no one ever dies
- Let $X(t)$ be population size at time t
- $\{X(t), t \geq 0\}$ is a pure birth process with birth rates:
  - $\lambda_n = n\lambda$ for all $n \geq 0$
  - called *Yule process*
- Consider a Yule process that starts with a single individual at time 0
- Let $T_i, i \geq 1$:
  - denote the time between $(i-1)$st and ith birth
  - i.e., time it takes for population size to go from i to $i+1$
- Distribution of $T_1, T_2, \ldots$:
  - $T_1, T_2, \ldots$ are independent and T_i is exponential with rate $i\lambda$ for all $i \geq 1$

Example: Yule Process (contd.)

- Recall:
  - T_i: time it takes for population size to go from i to $i+1$
  - $T_1, T_2, \ldots$ are independent and T_i is exponential with rate $i\lambda$ for all $i \geq 1$
- $P\{T_1 \leq t\}$:
  - $1 - e^{-\lambda t}$
- $P\{T_1 + T_2 \leq t\}$:
  - $\int_0^t P\{T_1 + T_2 \leq t | T_1 = x\} \lambda e^{-\lambda x} dx$
  - $\int_0^t \left(1 - e^{-2\lambda(t-x)}\right) \lambda e^{-\lambda x} dx$
  - $\left(1 - e^{-\lambda t}\right)^2$
- $P\{T_1 + T_2 + T_3 \leq t\}$:
  - $\int_0^t P\{T_1 + T_2 + T_3 \leq t | T_1 + T_2 = x\} dF_{T_1+T_2}(x)$
  - $\int_0^t \left(1 - e^{-3\lambda(t-x)}\right) 2\lambda e^{-\lambda x} \left(1 - e^{-\lambda x}\right) dx$
  - $\left(1 - e^{-\lambda t}\right)^3$
- In general, we can show by induction that:
  - $P\{T_1 + \cdots + T_j \leq t\} = \left(1 - e^{-\lambda t}\right)^j$
- For $j \geq 1$, we have that $P_{1j}(t)$:
  - $P\{X(t) = j | X(0) = 1\}$
  - $P\{X(t) \geq j | X(0) = 1\} - P\{X(t) \geq j+1 | X(0) = 1\}$
  - $P\{T_1 + \cdots + T_{j-1} \leq t\} - P\{T_1 + \cdots + T_j \leq t\}$
  - $\left(1 - e^{-\lambda t}\right)^{j-1} - \left(1 - e^{-\lambda t}\right)^j$
  - $e^{-\lambda t}\left(1 - e^{-\lambda t}\right)^{j-1}$
- Hence, starting with a single individual, population size at time t has geometric distribution with mean $e^{\lambda t}$

Example: Yule Process (contd.)

- Recall:
  - For $j \geq 1$, we have that $P_{1j}(t) = e^{-\lambda t}\left(1 - e^{-\lambda t}\right)^{j-1}$
  - Hence, starting with a single individual, population size at time t has geometric distribution with mean $e^{\lambda t}$
- Suppose the population starts with i individuals
- Population size at time t:
  - sum of i i.i.d geometric r.v.s
  - has a negative binomial distribution
- That is, for $j \geq i \geq 1$, we have that $P_{ij}(t)$:
  - $\binom{j-1}{i-1} e^{-\lambda t i}\left(1 - e^{-\lambda t}\right)^{j-i}$

Example: A Simple Epidemic Model

- Consider a population of m individuals that at time 0 consists of:
  - one "infected" individual,
  - $m-1$ "susceptible" individuals
- Once infected, an individual remains in that state forever
- In any small time interval h, any given infected person will cause, w.p. $\alpha h + o(h)$, any given susceptible to become infected
- Let $X(t)$ be the number of infected individuals in population at time t
- Then $\{X(t), t \geq 0\}$ is a pure birth process with birth rates:
  - $\lambda_n = \begin{cases} (m-n)n\alpha, & n \in \{1, \dots, m-1\}, \\ 0, & \text{otherwise.} \end{cases}$
  - this is because when there are n infected individuals, then each of the $m-n$ susceptibles will become infected at rate $n\alpha$
- Time until the total population is infected:
  - $T = \sum_{i=1}^{m-1} T_i$,
  - where T_i is the time to go from i infectives to $i+1$ infectives

Example: A Simple Epidemic Model (contd.)

- Recall:
  - ❑birth rates: $\lambda_n = \begin{cases} (m-n)n\alpha, & n \in \{1, \dots, m-1\}, \\ 0, & \text{otherwise.} \end{cases}$
  - ❑Time until the total population is infected: $T = \sum_{i=1}^{m-1} T_i$, where T_i is the time to go from i infectives to $i+1$ infectives
- Distribution of $T_1, \dots, T_{m-1}$:
  - ❑independent exponential r.v.s with rates $\lambda_i = (m-i)i\alpha$, for $i \in \{1, \dots, m-1\}$
- $E[T]$:
  - ❑$\frac{1}{\alpha} \sum_{i=1}^{m-1} \frac{1}{i(m-i)}$
- $\mathrm{var}(T)$:
  - ❑$\frac{1}{\alpha^2} \sum_{i=1}^{m-1} \left(\frac{1}{i(m-i)}\right)^2$

Kolmogorov Differential Equations

- Recall:
  - $P_{ij}(t) = P\{X(t+s) = j | X(s) = i\}$
- We will derive two sets of differential equations for $P_{ij}(t)$, which can sometimes be explicitly solved
- First, we prove some lemmas

Kolmogorov Differential Equations (contd.)

- Recall:
  - $P_{ij}(t) = P\{X(t+s) = j | X(s) = i\}$
- **Lemma 1**:

  a) $\lim_{t \to 0} \frac{1 - P_{ii}(t)}{t} = v_i$

  b) $\lim_{t \to 0} \frac{P_{ij}(t)}{t} = q_{ij}, i \neq j$
- **Proof**:
  - Since amount of time until a transition happens is exponentially distributed, it follows that the probability of two or more transitions in time t is $o(t)$
  - $1 - P_{ii}(t)$ equals the probability that a transition occurs within time t, plus a probability that is $o(t)$
  - So, $1 - P_{ii}(t) = v_i t + o(t)$, which implies part (a)
  - For $j \neq i$, $P_{ij}(t)$ equals the probability that a transition occurs in time t multiplied by the probability that this transition is into state j, plus a probability that is $o(t)$
  - So, $P_{ij}(t) = t v_i P_{ij} + o(t)$, which implies part (b)
- **Lemma 2**: For all s, t:
  - $P_{ij}(t+s) = \sum_{k=0}^{\infty} P_{ik}(t) P_{kj}(s)$
- **Proof**:
  - By conditioning on value of $X(t)$ and using Markovian property, we get $P_{ij}(t+s)$:
    - $\sum_{k=0}^{\infty} P\{X(t+s) = j \,|X(t) = k, X(0) = i\} P\{X(t) = k | X(0) = i\}$
    - $\sum_{k=0}^{\infty} P\{X(t+s) = j \,|X(t) = k\} P\{X(t) = k | X(0) = i\}$
    - $\sum_{k=0}^{\infty} P_{kj}(s)\, P_{ik}(t)$
- Note that Lemma 2 is the continuous-time version of the Chapman-Kolmogorov equations of DTMCs

Kolmogorov Differential Equations (contd.)

- Recall:
  - **Lemma 1**:
    - *a)* $\lim_{t\to 0} \frac{1-P_{ii}(t)}{t} = v_i$
    - *b)* $\lim_{t\to 0} \frac{P_{ij}(t)}{t} = q_{ij}, i \neq j$
  - **Lemma 2**: For all s, t:
    - $P_{ij}(t+s) = \sum_{k=0}^{\infty} P_{ik}(t)P_{kj}(s)$
- From Lemma 2, we get:
  - $P_{ij}(t+h) = \sum_{k=0}^{\infty} P_{ik}(h)P_{kj}(t)$
- So, $P_{ij}(t+h) - P_{ij}(t)$:
  - $\sum_{k\neq i} P_{ik}(h)P_{kj}(t) - [1 - P_{ii}(h)]P_{ij}(t)$
- Dividing by h, taking limits as $h \to 0$, and using Lemma 1, we get:

1) $\lim_{h\to 0} \frac{P_{ij}(t+h)-P_{ij}(t)}{h} = \lim_{h\to 0} \sum_{k\neq i} \frac{P_{ik}(h)}{h} P_{kj}(t) - v_i P_{ij}(t)$

- Assuming that we can justify interchange of the limit and summation on RHS of 1) (proof omitted) and using Lemma 1, we get following result
- **Theorem (Kolmogorov's Backward Equations)**: For all i, j, and $t \geq 0$:
  - $P'_{ij}(t) = \sum_{k\neq i} q_{ik} P_{kj}(t) - v_i\, P_{ij}(t)$

Kolmogorov Differential Equations (contd.)

- Recall:
  - **Theorem (Kolmogorov's Backward Equations)**: For all i, j, and $t \geq 0$:
    - $P'_{ij}(t) = \sum_{k \neq i} q_{ik} P_{kj}(t) - v_i \, P_{ij}(t)$
- Above equations called backward equations because in computing probability distribution of state at time $t + h$:
  - we conditioned on the state back at time h
- That is, we started our calculation with:
  - $P_{ij}(t + h) = \sum_{k=0}^{\infty} P_{ik}(h) P_{kj}(t)$
- Next, we derive another set of equations, called *Kolmogorov's forward equations*, by conditioning on state at time t to get:
  - $P_{ij}(t + h) = \sum_{k=0}^{\infty} P_{ik}(t) P_{kj}(h)$
- So, $P_{ij}(t + h) - P_{ij}(t)$:
  - $\sum_{k=0}^{\infty} P_{ik}(t) P_{kj}(h) - P_{ij}(t)$
  - $\sum_{k \neq j} P_{ik}(t) P_{kj}(h) - \left[1 - P_{jj}(h)\right] P_{ij}(t)$
- Hence, $\lim_{h \to 0} \frac{P_{ij}(t+h) - P_{ij}(t)}{h}$:
  - $\lim_{h \to 0} \left\{ \sum_{k \neq j} P_{ik}(t) \frac{P_{kj}(h)}{h} - \frac{1 - P_{jj}(h)}{h} P_{ij}(t) \right\}$

Kolmogorov Differential Equations (contd.)

- Recall:
  - **Lemma 1**:

    a) $\lim_{t\to 0} \frac{1-P_{ii}(t)}{t} = v_i$

    b) $\lim_{t\to 0} \frac{P_{ij}(t)}{t} = q_{ij},\ i \neq j$

  1) $\lim_{h\to 0} \frac{P_{ij}(t+h)-P_{ij}(t)}{h} = \lim_{h\to 0} \left\{ \sum_{k\neq j} P_{ik}(t) \frac{P_{kj}(h)}{h} - \frac{1-P_{jj}(h)}{h} P_{ij}(t) \right\}$

- Assuming that we can exchange limit with summation in RHS of 1) and using Lemma 1, we get:

  2) $P'_{ij}(t) = \sum_{k\neq j} q_{kj} P_{ik}(t) - v_j\ P_{ij}(t)$

- However, we cannot always justify above exchange of limit and summation:
  - hence, 2) is not always valid
- However, it can be shown that 2) is valid for most models:
  - including all birth and death processes and all finite-state models
- **Theorem (Kolmogorov's Forward Equations)**: Under suitable conditions, for all i, j, and $t \geq 0$:
  - $P'_{ij}(t) = \sum_{k\neq j} q_{kj} P_{ik}(t) - v_j\ P_{ij}(t)$
- We omit a discussion of the exact conditions under which Kolmogorov's forward equations can be shown to be valid

Example 1

- Recall: Kolmogorov's Forward Equations:
  - $P'_{ij}(t) = \sum_{k \neq j} q_{kj} P_{ik}(t) - v_j \, P_{ij}(t)$
- Consider a two-state CTMC, which:
  - spends an exponential time with rate λ in state 0 before going to state 1,
  - spends an exponential time with rate μ in state 1 before going to state 0
- By Kolmogorov's forward equations:
  - $P'_{00}(t) = \mu P_{01}(t) - \lambda P_{00}(t)$
  - $\qquad\qquad = -(\lambda + \mu) P_{00}(t) + \mu$
- Hence:
  - $e^{(\lambda+\mu)t}[P'_{00}(t) + (\lambda + \mu) P_{00}(t)] = \mu e^{(\lambda+\mu)t}$
  - i.e., $\frac{d}{dt}\left[e^{(\lambda+\mu)t} P_{00}(t)\right] = \mu e^{(\lambda+\mu)t}$
- Integrating and using the fact that $P_{00}(0) = 1$, we get:
  - $P_{00}(t) = \frac{\mu}{\lambda+\mu} + \frac{\lambda}{\lambda+\mu} e^{-(\lambda+\mu)t}$
- By symmetry:
  - $P_{11}(t) = \frac{\lambda}{\lambda+\mu} + \frac{\mu}{\lambda+\mu} e^{-(\lambda+\mu)t}$

Example 2

- Recall:
  - ❑Kolmogorov's Forward Equations:
    - $P'_{ij}(t) = \sum_{k \neq j} q_{kj} P_{ik}(t) - v_j \, P_{ij}(t)$
  - ❑for a birth and death process:
    - birth rates: $\lambda_i = q_{i,i+1}$ for $i \geq 0$
    - death rates: $\mu_i = q_{i,i-1}$ for $i \geq 1$
- Kolmogorov's forward equations:
  - ❑$P'_{i0}(t) = \mu_1 P_{i1}(t) - \lambda_0 P_{i0}(t)$,
  - ❑$P'_{ij}(t) = \lambda_{j-1} P_{i,j-1}(t) + \mu_{j+1} P_{i,j+1}(t) - \left(\lambda_j + \mu_j\right) P_{i,j}(t)$, for $j \neq 0$

Example 3

- Recall:
  - Kolmogorov's Forward Equations:
    - $P'_{ij}(t) = \sum_{k \neq j} q_{kj} P_{ik}(t) - v_j \, P_{ij}(t)$
  - for a pure birth process:
    - birth rates: $\lambda_i = q_{i,i+1}$ for $i \geq 0$
- Kolmogorov's forward equations:

  1) $P'_{ii}(t) = -\lambda_i P_{ii}(t)$,

  2) $P'_{ij}(t) = \lambda_{j-1} P_{i,j-1}(t) - \lambda_j P_{ij}(t)$, for $j > i$
- Integrating 1) and using $P_{ii}(0) = 1$, we get:
  - $P_{ii}(t) = e^{-\lambda_i t}$,
  - which we already know since time until transition from state i to $i+1$ is exponential with rate λ_i
- By 2), for $j > i$, we get:
  - $e^{\lambda_j t} \, \lambda_{j-1} P_{i,j-1}(t) = e^{\lambda_j t}\left[P'_{ij}(t) + \lambda_j P_{ij}(t)\right]$
  - $= \frac{d}{dt}\left[e^{\lambda_j t} P_{ij}(t)\right]$
- Integrating and using $P_{ij}(0) = 0$, we get the recursion:

  3) $P_{ij}(t) = \lambda_{j-1} e^{-\lambda_j t} \int_0^t e^{\lambda_j s} P_{i,j-1}(s) ds$, for $j > i$
- In the special case where the process is a Yule process, $\lambda_j = j\lambda$, we can use 3) to verify the following result that we earlier derived:
  - $P_{ij}(t) = \binom{j-1}{i-1} e^{-\lambda t i}\left(1 - e^{-\lambda t}\right)^{j-i}$, for $j \geq i \geq 1$

Example 4

- Recall:
  - ❑Kolmogorov's Backward Equations:
    - $P'_{ij}(t) = \sum_{k \neq i} q_{ik} P_{kj}(t) - v_i \, P_{ij}(t)$
  - ❑for a birth and death process:
    - birth rates: $\lambda_i = q_{i,i+1}$ for $i \geq 0$
    - death rates: $\mu_i = q_{i,i-1}$ for $i \geq 1$
  - ❑for a pure birth process:
    - birth rates: $\lambda_i = q_{i,i+1}$ for $i \geq 0$
- Kolmogorov's backward equations for a pure birth process:
  - ❑$P'_{ij}(t) = \lambda_i P_{i+1,j}(t) - \lambda_i P_{ij}(t)$, for $i \geq 0$
- Kolmogorov's backward equations for a birth and death process:
  - ❑$P'_{0j}(t) = \lambda_0 P_{1,j}(t) - \lambda_0 P_{0j}(t)$,
  - ❑$P'_{ij}(t) = \lambda_i P_{i+1,j}(t) + \mu_i P_{i-1,j}(t) - (\lambda_i + \mu_i) P_{i,j}(t)$, for $i > 0$

Example 5

- Recall: Kolmogorov's backward equations for a birth and death process:
  1) $P'_{0j}(t) = \lambda_0\left[P_{1,j}(t) - P_{0j}(t)\right]$,
  2) $P'_{ij}(t) = \lambda_i P_{i+1,j}(t) + \mu_i P_{i-1,j}(t) - (\lambda_i + \mu_i)P_{i,j}(t)$, for $i > 0$
- Consider a machine:
  - which works for an exponential amount of time with rate λ before breaking down,
  - it takes an exponential time with rate μ to repair the machine
- If machine is in working condition at time 0, then what is the probability that it will be working at time $t = 10$?
- Let machine in working condition (respectively, repair) be in state 0 (respectively, 1)
- Process is a birth and death process
- By 1) and 2), we get:
  3) $P'_{00}(t) = \lambda[P_{10}(t) - P_{00}(t)]$,
  4) $P'_{10}(t) = \mu P_{00}(t) - \mu P_{10}(t)$
- Multiplying 3) by μ and 4) by λ, adding the two resulting equations, and integrating the result gives:
  - $\mu P_{00}(t) + \lambda P_{10}(t) = c$
- Since $P_{00}(0) = 1$ and $P_{10}(0) = 0$, we get $c = \mu$, and hence:
  5) $\mu P_{00}(t) + \lambda P_{10}(t) = \mu$
- By 3) and 5), we get:
  6) $P'_{00}(t) = \mu - (\lambda + \mu)P_{00}(t)$
- Equation 6) is the same equation that we solved in Example 1 above to get:
  7) $P_{00}(t) = \frac{\mu}{\lambda+\mu} + \frac{\lambda}{\lambda+\mu}e^{-(\lambda+\mu)t}$
- By (5) and (7):
  - $P_{10}(t) = \frac{\mu}{\lambda+\mu} - \frac{\mu}{\lambda+\mu}e^{-(\lambda+\mu)t}$
- Required probability:
  - $P_{00}(10) = \frac{\mu}{\lambda+\mu} + \frac{\lambda}{\lambda+\mu}e^{-10(\lambda+\mu)}$

Matrix Representation of Kolmogorov's Differential Equations

- Recall:
  - Kolmogorov's backward equations:
    - $P'_{ij}(t) = \sum_{k \neq i} q_{ik} P_{kj}(t) - v_i \, P_{ij}(t)$
  - Kolmogorov's forward equations:
    - $P'_{ij}(t) = \sum_{k \neq j} q_{kj} P_{ik}(t) - v_j \, P_{ij}(t)$
- Let:
  - $r_{ij} = \begin{cases} q_{ij}, & \text{if } i \neq j, \\ -v_i, & \text{if } i = j. \end{cases}$
- r_{ij} are known as *conditional rates*, i.e., they are rates conditioned on CTMC being in state i
- Then Kolmogorov's backward equations can be written as:
  - $P'_{ij}(t) = \sum_k r_{ik} P_{kj}(t)$
- Also, Kolmogorov's forward equations can be written as:
  - $P'_{ij}(t) = \sum_k r_{kj} P_{ik}(t)$
- Let $R, P(t)$, and $P'(t)$ be the matrices whose (i,j)th element is r_{ij}, $P_{ij}(t)$, and $P'_{ij}(t)$, respectively
- R is known as the *conditional-rate matrix*
- Note that the row sums of R all equal 0
- Then Kolmogorov's backward and forward equations can be respectively written as:

  1) $P'(t) = RP(t)$

  2) $P'(t) = P(t)R$
- Equations 1) and 2) suggest the solution:

  3) $P(t) = e^{Rt} \equiv \sum_{t=0}^{\infty} \frac{(Rt)^i}{i!}$
- It can be shown that 3) is valid provided that v_i are bounded
- However, use of 3) to compute $P(t)$ turns out to be inefficient

CTMC and Its Embedded DTMC

- Consider a CTMC and let P_{ij} be the transition probabilities of the embedded DTMC
- For any $i \neq j$, we say that state i can reach state j in the CTMC if for some $t \geq 0$, $P_{ij}(t) > 0$
- Clearly, i can reach j in the CTMC iff there is a path of positive probability from i to j in the embedded DTMC, i.e., i can reach state j in the embedded DTMC
- For CTMCs, define communication classes and the terms open, closed, and irreducible as we did for DTMCs
- **Theorem**:
  - A CTMC and its embedded DTMC have the same communication classes
  - In particular, the CTMC is irreducible iff the embedded DTMC is irreducible
- **Note**:
  - Since sojourn times can be arbitrarily small, it is easy to see that if i can reach j, then $P_{ij}(t) > 0$ for all $t > 0$

Limiting Probabilities

- Recall:
  - **Theorem**: Suppose a semi-Markov process is irreducible and T_{ii} has a non-lattice CDF with finite mean. Also, suppose the embedded DTMC $\{X_n, n \geq 0\}$ is positive recurrent. Then:
    - $P_i = \frac{\pi_i \mu_i}{\sum_k \pi_k \mu_k}$
  - a CTMC is a semi-Markov process with:
    - $F_{ij}(t) = 1 - e^{-v_i t}$
- Let P_{ij} be the transition probabilities of the embedded DTMC
- If the embedded DTMC is irreducible and positive-recurrent, then from above theorem, it follows that the limiting probabilities $P_j = \lim_{t \to \infty} P_{ij}(t)$ are given by:

  1) $P_j = \frac{\pi_j / v_j}{\sum_i \pi_i / v_i}$,
  - where the π_i are the stationary probabilities, i.e., the unique non-negative solution of following system:

    2) $\pi_j = \sum_i \pi_i P_{ij}$

    3) $\sum_i \pi_i = 1$
- By 1), 2), and 3), we see that the P_j are the unique non-negative solution of following system:

  4) $v_j P_j = \sum_i v_i P_i P_{ij}$,

  5) $\sum_j P_j = 1$
- Using $q_{ij} = v_i P_{ij}$, 4) and 5) can be written as:

  6) $v_j P_j = \sum_i P_i q_{ij}$,

  7) $\sum_j P_j = 1$
- By the results we proved for semi-Markov processes, it follows that P_j also equals the long-run proportion of time that the process is in state j

Limiting Probabilities (contd.)

- Recall: P_j are the unique non-negative solution of following system:

  6) $v_j P_j = \sum_i P_i q_{ij}$,

  7) $\sum_j P_j = 1$

- Interpretation:
  - ❑ LHS of 6) = rate at which process makes transitions out of state j
  - ❑ RHS of 6) = rate at which process makes transitions into state j
- Since 6) balances (equates) the incoming and outgoing transition rates, 6) are often referred to as *balance equations*
- A generalization of 6): If A is any subset of the set of states, then:

  8) $\sum_{i \in A} \sum_{j \in A^c} P_i q_{ij} = \sum_{i \in A^c} \sum_{j \in A} P_i q_{ij}$

- Equation 8) follows from corresponding equation for DTMCs, which we showed earlier
- By choosing A appropriately, we can often use 8) to find $\{P_j\}$ quickly
- E.g., for the M/M/1 queue, we can select $A = \{0, \dots, j\}$ and $A^c = \{j+1, j+2, \dots\}$ for any state j

Example

- Recall the shoe shine shop example we discussed earlier:
  - there is a shoe shine shop with two chairs, 1 and 2
  - process is a CTMC with following states:
    - 0: system is empty
    - 1: customer in chair 1
    - 2: customer in chair 2
  - Transition rates:
    - $v_0 = \lambda$, $v_1 = \mu_1$, and $v_2 = \mu_2$
- Balance equations:
  - $\lambda P_0 = \mu_2 P_2$
  - $\mu_1 P_1 = \lambda P_0$
  - $\mu_2 P_2 = \mu_1 P_1$
- Solving balance equations along with $P_0 + P_1 + P_2 = 1$, we get:
  - $P_0 = \frac{\mu_1 \mu_2}{\mu_1 \mu_2 + \lambda(\mu_1 + \mu_2)}$,
  - $P_1 = \frac{\lambda \mu_2}{\mu_1 \mu_2 + \lambda(\mu_1 + \mu_2)}$,
  - $P_2 = \frac{\lambda \mu_1}{\mu_1 \mu_2 + \lambda(\mu_1 + \mu_2)}$

Limiting Probabilities (contd.)

- Recall: for an irreducible CTMC, P_j are the unique non-negative solution of following system:

  6) $v_j P_j = \sum_i P_i q_{ij}$,

  7) $\sum_j P_j = 1$

- If initial state of CTMC is chosen according to the limiting probabilities $\{P_j\}$, then:
  - resultant process will be stationary,
  - i.e., the probability, $P(X(t) = j)$, of process being in state j at time t is P_j for all t and j
- That is:

  8) $P(X(t) = j) = P_j$, for all j and t

- **Proof of 8)**:
  - $P(X(t) = j) = \sum_i P(X(0) = i) P(X(t) = j | X(0) = i)$
  - $= \sum_i P_i P_{ij}(t)$
  - $= \sum_i P_{ij}(t) \lim_{s \to \infty} P_{ki}(s)$
  - $= \lim_{s \to \infty} \sum_i P_{ij}(t) P_{ki}(s)$
  - $= \lim_{s \to \infty} P_{kj}(t + s)$
  - $= P_j$,
  - where interchange of limit and infinite summation is justified because of bounded convergence theorem
- We have shown that the limiting probabilities $\{P_j\}$ satisfy the following equations:

  9) $\sum_i P_i P_{ij}(t) = P_j$

Limiting Probabilities (contd.)

- Recall: for an irreducible CTMC, P_j are the unique non-negative solution of following system:

  6) $v_j P_j = \sum_i P_i q_{ij}$,

  7) $\sum_j P_j = 1$

- On previous slide, we showed that if a probability distribution $\{P_j\}$ satisfies the balance equations 6), then it also satisfies the following equations:

  9) $\sum_i P_i P_{ij}(t) = P_j$

- Now, by 7), $P_i > 0$ for some i; so, 9) implies that $P_j > 0$ for every state j
- When the CTMC is irreducible and $P_j > 0$ for all j, the CTMC is said to be *ergodic*
- **Note**: We also refer to P_j as *steady-state* probabilities and $\{P_j\}$ as the steady-state distribution

Limiting Probabilities for a Birth and Death Process

- Recall:
  - P_j are the unique non-negative solution of following system:
    1) $v_j P_j = \sum_i P_i q_{ij}$,
    2) $\sum_j P_j = 1$
  - If A is any subset of the set of states, then:
    3) $\sum_{i \in A} \sum_{j \in A^c} P_i q_{ij} = \sum_{i \in A^c} \sum_{j \in A} P_i q_{ij}$
  - For a birth and death process:
    - birth rates: $\lambda_i = q_{i,i+1}$ for $i \geq 0$
    - death rates: $\mu_i = q_{i,i-1}$ for $i \geq 1$
- By 1), for $j = 0$ and $j = n > 0$, we get:
  4) $\lambda_0 P_0 = \mu_1 P_1$
  5) $(\lambda_n + \mu_n) P_n = \mu_{n+1} P_{n+1} + \lambda_{n-1} P_{n-1}$
- Simplifying 4) and 5), or directly by using 3), we get:
  6) $\lambda_n P_n = \mu_{n+1} P_{n+1}$, for $n \geq 0$
- By 6), we get:
  - $P_1 = \frac{\lambda_0}{\mu_1} P_0$,
  - $P_2 = \frac{\lambda_1}{\mu_2} P_1 = \frac{\lambda_1 \lambda_0}{\mu_2 \mu_1} P_0$,
  - $P_3 = \frac{\lambda_2}{\mu_3} P_2 = \frac{\lambda_2 \lambda_1 \lambda_0}{\mu_3 \mu_2 \mu_1} P_0$
  - $\vdots$
  - $P_n = \frac{\lambda_{n-1}}{\mu_n} P_{n-1} = \frac{\lambda_{n-1} \lambda_{n-2} \dots \lambda_0}{\mu_n \mu_{n-1} \dots \mu_1} P_0$
- Substituting these probabilities into 2), we get:
  - $P_0 = \frac{1}{1 + \sum_{n=1}^{\infty} \frac{\lambda_{n-1} \lambda_{n-2} \dots \lambda_0}{\mu_n \mu_{n-1} \dots \mu_1}}$
  - $P_n = \frac{\lambda_{n-1} \lambda_{n-2} \dots \lambda_0}{\mu_n \mu_{n-1} \dots \mu_1 \left(1 + \sum_{n=1}^{\infty} \frac{\lambda_{n-1} \lambda_{n-2} \dots \lambda_0}{\mu_n \mu_{n-1} \dots \mu_1}\right)}$, for all $n \geq 1$
- Condition needed for limiting probabilities to exist:
  - $\sum_{n=1}^{\infty} \frac{\lambda_{n-1} \lambda_{n-2} \dots \lambda_0}{\mu_n \mu_{n-1} \dots \mu_1} < \infty$

Example 1

- Recall: for a birth and death process:

  1) $P_0 = \frac{1}{1+\sum_{n=1}^{\infty}\frac{\lambda_{n-1}\lambda_{n-2}\ldots\lambda_0}{\mu_n\mu_{n-1}\ldots\mu_1}}$

  2) $P_n = \frac{\lambda_{n-1}\lambda_{n-2}\ldots\lambda_0}{\mu_n\mu_{n-1}\ldots\mu_1\left(1+\sum_{n=1}^{\infty}\frac{\lambda_{n-1}\lambda_{n-2}\ldots\lambda_0}{\mu_n\mu_{n-1}\ldots\mu_1}\right)}$, for all $n \geq 1$

  - ❑Condition needed for limiting probabilities to exist:

    3) $\sum_{n=1}^{\infty}\frac{\lambda_{n-1}\lambda_{n-2}\ldots\lambda_0}{\mu_n\mu_{n-1}\ldots\mu_1} < \infty$

- For the CTMC of M/M/1 queue, which we discussed earlier, $\lambda_n = \lambda$ and $\mu_n = \mu$, for all n
- So, by 1), 2), and 3), P_n:
  - ❑$\left(\frac{\lambda}{\mu}\right)^n\left(1-\frac{\lambda}{\mu}\right)$, for all $n \geq 0$,
  - ❑provided that $\frac{\lambda}{\mu} < 1$

Example 2

- Recall: for a birth and death process:

  1) $P_0 = \frac{1}{1+\sum_{n=1}^{\infty}\frac{\lambda_{n-1}\lambda_{n-2}\dots\lambda_0}{\mu_n\mu_{n-1}\dots\mu_1}}$

  2) $P_n = \frac{\lambda_{n-1}\lambda_{n-2}\dots\lambda_0}{\mu_n\mu_{n-1}\dots\mu_1\left(1+\sum_{n=1}^{\infty}\frac{\lambda_{n-1}\lambda_{n-2}\dots\lambda_0}{\mu_n\mu_{n-1}\dots\mu_1}\right)}$, for all $n \geq 1$

- Consider a job shop consisting of M machines and a single repairman
- Amount of time:
  - ❑ a machine runs before breaking down is exponentially distributed with rate λ,
  - ❑ it takes repairman to fix any broken machine is exponentially distributed with rate μ
- Suppose the state is:
  - ❑ n whenever n machines are down
- This is a birth and death process with parameters:
  - ❑ $\mu_n = \mu$, for $n \geq 1$,
  - ❑ $\lambda_n = \begin{cases}(M-n)\lambda, & n \leq M,\\ 0, & n > M\end{cases}$

- From 1) and 2), it follows that P_n, the limiting probability that n machines will not be in use is given by:

  1) $P_0 = \frac{1}{1+\sum_{k=1}^{M}\left(\frac{\lambda}{\mu}\right)^k\frac{M!}{(M-k)!}}$

  2) $P_n = \frac{\left(\frac{\lambda}{\mu}\right)^n\frac{M!}{(M-n)!}}{1+\sum_{k=1}^{M}\left(\frac{\lambda}{\mu}\right)^k\frac{M!}{(M-k)!}}$, for all $n \in \{1, \dots, M\}$

- Average number of machines not in use:
  - ❑ $\sum_{n=0}^{M} nP_n = \frac{\sum_{n=0}^{M} n\left(\frac{\lambda}{\mu}\right)^n\frac{M!}{(M-n)!}}{1+\sum_{k=1}^{M}\left(\frac{\lambda}{\mu}\right)^k\frac{M!}{(M-k)!}}$
- Long-run proportion of time that a given machine is working:
  - ❑ $\sum_{n=0}^{M} P\{\text{machine is working}|n \text{ not working}\}\, P_n$
  - ❑ $\sum_{n=0}^{M} \frac{M-n}{M} P_n$

Time Reversibility

- Consider an ergodic CTMC, whose initial state is chosen according to the stationary probabilities P_j
- Such a process is stationary
- Consider this process going backward in time
- By Markovian property of original CTMC, for $s > 0$:
  - $P\{X(t-s) = j | X(t) = i, X(y), y > t\} = P\{X(t-s) = j | X(t) = i\}$
- So, reverse process is also a CTMC
- **Proposition**: Amount of time the reverse CTMC spends in state i on a visit is exponential with same rate v_i as in forward CTMC
- Intuitively, this is because amount of time spent in a state is the same whether one is going forward or backward in time
- **Proof**:
  - Suppose reverse CTMC is in state i at time t
  - Then $P\{\text{process is in state } i \text{ throughout } [t-s, t] | X(t) = i\}$:
    - $\frac{P\{\text{process is in state } i \text{ throughout } [t-s,t]\}}{P\{X(t)=i\}}$
    - $\frac{P\{X(t-s)=i\}e^{-v_i s}}{P\{X(t)=i\}}$
    - $e^{-v_i s}$

Time Reversibility (contd.)

- Recall:
  - going backwards in time, the amount of time that the CTMC spends in state i during a visit is exponential with same rate v_i as forward CTMC
- Also, as shown during our discussion of time reversible DTMCs:
  - the sequence of states visited by reverse process constitutes a DTMC with transition probabilities:
    - $P_{ij}^* = \frac{\pi_j P_{ji}}{\pi_i}$,
    - where $\{\pi_j, j \geq 0\}$ are stationary probabilities of the embedded DTMC of forward process, which has transition probabilities P_{ij}

Time Reversibility (contd.)

- Recall: reverse process is a CTMC with:
  - ❑ transition rate out of state i equal to v_i,
  - ❑ transition probabilities of embedded DTMC equal to $P_{ij}^* = \frac{\pi_j P_{ji}}{\pi_i}$
- Let $q_{ij}^* = v_i P_{ij}^*$ be transition rate from state i to state j of reverse CTMC
- **Proposition**: $P_i q_{ij}^* = P_j q_{ji}$, for all states i and j
- **Proof**:
  - ❑ We know that $P_k = \frac{\pi_k / v_k}{C}$, where $C = \sum_i \frac{\pi_i}{v_i}$

  1) So, $\frac{\pi_j}{\pi_i} = \frac{v_j P_j}{v_i P_i}$

  2) Now, $q_{ij}^* = v_i P_{ij}^* = \frac{v_i \pi_j P_{ji}}{\pi_i}$

  - ❑ By 1) and 2), we get:
    - ○ $q_{ij}^* = \frac{v_j P_j P_{ji}}{P_i} = \frac{P_j q_{ji}}{P_i}$

Time Reversibility (contd.)

- Recall:
  - $q_{ij}^* = v_i P_{ij}^*$ is transition rate from state i to state j of reverse CTMC
  - **Proposition**: $P_i q_{ij}^* = P_j q_{ji}$, for all states i and j
- **Proposition**: $\{P_j\}$ are the stationary probabilities of the reverse CTMC
- Intuitively, this is because:
  - P_j is the proportion of time that the process is in state j when looking in forward direction of time,
  - Hence, it is also the proportion of time that the process is in state j when looking in backward direction
- **Proof**:
  - Summing the equation $P_i q_{ij}^* = P_j q_{ji}$ over all states i, we get:
    - $\sum_i P_i q_{ij}^* = P_j v_j$
  - So, $\{P_j\}$ satisfy the balance equations of the reverse CTMC and hence are the stationary probabilities of reverse CTMC

Time Reversibility (contd.)

- Recall:
  - **Proposition**: $P_i q_{ij}^* = P_j q_{ji}$, for all states i and j
  - **Proposition**: $\{P_j\}$ are the stationary probabilities of the reverse CTMC
- Interpretation of $P_i q_{ij}^* = P_j q_{ji}$:
  - LHS is rate at which reverse CTMC makes a transition from i to j
  - RHS is rate at which forward CTMC makes a transition from j to i
  - These rates are equal because every time the CTMC makes a transition from j to i in forward direction, the process when looked in reverse direction is making a transition from i to j

Time Reversibility (contd.)

- Recall:
  - **Proposition**: $P_i q_{ij}^* = P_j q_{ji}$, for all states i and j
  - **Proposition**: $\{P_j\}$ are the stationary probabilities of the reverse CTMC
- **Definition**: The stationary CTMC is said to be time reversible if for all i and j:

  1) $q_{ij}^* = q_{ij}$
- The condition in 1) is equivalent to:

  2) $P_i q_{ij} = P_j q_{ji}$
- Interpretation:
  - rate at which process goes directly from state i to state j is equal to rate at which it goes directly from j to i
- **Proposition**: A stationary ergodic birth and death process is time reversible
- **Proof**:
  - Recall that if A is any subset of the set of states, then:
    - $\sum_{i \in A} \sum_{j \in A^c} P_i q_{ij} = \sum_{i \in A^c} \sum_{j \in A} P_i q_{ij}$
  - So, $P_n \lambda_n = P_{n+1} \mu_{n+1}$

Time Reversibility (contd.)

- Recall:
  - The stationary CTMC is said to be time reversible if for all i and j:
    1) $q_{ij}^* = q_{ij}$
  - The condition in 1) is equivalent to:
    2) $P_i q_{ij} = P_j q_{ji}$
- **Proposition**: If $x_j \geq 0$, for all $j \in \{0,1,2, \dots\}$, and:

  3) $\sum_j x_j = 1$

  4) $x_i q_{ij} = x_j q_{ji}$, for all i, j,

  then the stationary CTMC is time reversible and the x_j's are the stationary probabilities
- **Proof**:
  - Summing 4) over all i gives:
    - $\sum_i x_i q_{ij} = x_j \sum_i q_{ji} = x_i \, v_i$
  - Thus, the x_j's satisfy the balance equations and hence are the stationary probabilities
  - By 4), the CTMC is time reversible

Truncated Time-Reversible CTMC

- Consider a CTMC whose state space is S
- The CTMC is said to be *truncated* to set $A \subseteq S$ if:
  - ❑ q_{ij} is changed to 0 for all $i \in A$ and $j \notin A$
  - ❑ all other q_{ij} remain the same
- Hence, transitions out of the set of states A are not allowed
- **Proposition**: A time-reversible CTMC with limiting probabilities P_j, $j \in S$, that is truncated to the set $A \subseteq S$ and remains irreducible is also time reversible and has limiting probabilities:

  1) $P_j^A = \frac{P_j}{\sum_{k \in A} P_k}$, for $j \in A$
- **Proof**:
  - ❑ It suffices to show that:

    2) $P_i^A q_{ij} = P_j^A q_{ji}$, for all $i \in A$ and $j \in A$
  - ❑ By 1), 2) is equivalent to:

    3) $P_i q_{ij} = P_j q_{ji}$, for all $i \in A$ and $j \in A$
  - ❑ However, 3) is true because original CTMC is time reversible

Example

- Recall: in an earlier example, we showed that for CTMC of M/M/1 queue, stationary probabilities are given by:

  1) $P_n = \left(\frac{\lambda}{\mu}\right)^n \left(1 - \frac{\lambda}{\mu}\right)$, for all $n \geq 0$,

  - provided that $\frac{\lambda}{\mu} < 1$
- Now, consider an M/M/1/N queue:
  - it is an M/M/1 queue with the change that arrivals finding N customers in the system do not enter, but rather are lost
- The M/M/1/N queue can be regarded as truncated version of M/M/1 queue
- Hence, by 1), it is time reversible with limiting probabilities given by:
  - $P_j = \frac{\left(\frac{\lambda}{\mu}\right)^j}{\sum_{i=0}^{N}\left(\frac{\lambda}{\mu}\right)^i}$, for $j \in \{0, \dots, N\}$

Recurrence

- For all $t \geq 0$, let:
  - $Y(t) = \inf\{s > 0: X(t+s) \neq X(t)\}$
- That is, $Y(t)$ is remaining time that process spends in the state that it is in at time t
- For a CTMC $X(t)$, $t \geq 0$, on state space S, for $j \in S$, and with $X(0) = j$, let:
  - $S_{jj} = \inf\{t \geq 0: t > Y(0), X(t) = j\}$
- That is, S_{jj} is the time to return to state j after once leaving j
- **Definition**:
  - State $j \in S$ is said to be *recurrent* in the CTMC if $P\left(S_{jj} < \infty\right) = 1$
  - An irreducible CTMC is called recurrent if each state $j \in S$ is recurrent
- **Definition**:
  - Suppose state $j \in S$ of the CTMC is recurrent
  - It is said to be *positive recurrent* if $E\left(S_{jj}\right) < \infty$ and *null recurrent* if $E\left(S_{jj}\right) = \infty$

Periodicity

- Recall: period of a state i of a DTMC is:
  - $d(i) = \gcd\{n > 0: P_{ii}^n > 0\}$
- In a CTMC, for every $i \in S$ and $t > 0$:
  - $P_{ii}(t) \geq e^{-v_i t} > 0$
- So, there is no basic time-step in multiples of which the CTMC is found in the state i, having started in state i
- Hence, there is no notion of periodicity for CTMCs
- However, it is easy to see that a CTMC can have a periodic embedded DTMC:
  - e.g., period of embedded DTMC of the CTMC of an M/M/s queue is 2

Recurrence of CTMC and Its Embedded DTMC

- **Theorem**: An irreducible CTMC is recurrent iff its embedded DTMC is recurrent
- **Proof**:
  - Result trivially true if $|S| = 1$; so, assume that $|S| \geq 2$
  - Suppose the embedded DTMC, denoted by X_k, is recurrent
  - Then for every state $j \in S$, the first return time T_{jj} is finite w.p. 1, i.e., $P\left(T_{jj} < \infty\right) = 1$
  - Since X_k, $k \geq 0$, is recurrent, there cannot be any absorbing states
  - Hence, for all $i \in S$, $0 < v_i < \infty$
  - If $T_{jj} < \infty$, then S_{jj} is the sum of a finite number of exponentially distributed state sojourn times, each with finite mean:
    - so, S_{jj} is finite w.p. 1
  - Thus, for each state j, $P\left(S_{jj} < \infty\right) = 1$:
    - hence, the CTMC is recurrent
  - Similarly, if the embedded DTMC is transient, then there is a state $j \in S$ such that $P\left(T_{jj} = \infty\right) > 0$
  - Then $P\left(S_{jj} = \infty\right) > 0$, and the CTMC is also transient

Recurrence of CTMC and Its Embedded DTMC (contd.)

- Recall:
  - **Theorem**: An irreducible CTMC is recurrent iff its embedded DTMC is recurrent
- Above theorem shows that recurrence/ transience of an irreducible CTMC can be inferred from the same properties of its irreducible embedded DTMC
- However, in general, a CTMC and its embedded DTMC do not share the properties of positive recurrence and null recurrence, e.g.:
  - It is possible that a CTMC is positive recurrent, while its embedded DTMC is null recurrent, and vice versa

Solutions of $\pi = \pi P$

- Consider an irreducible DTMC
- We showed that if it is positive recurrent, then there is a unique probability vector π satisfying $\pi = \pi P$
- Does the system of equations $\pi = \pi P$ have a non-zero solution π even if the DTMC is not positive recurrent?
- If it is null recurrent, then:
  - ❑A non-zero solution to the system of equations $\pi = \pi P$ with $\pi_i \geq 0, \forall i$ exists
  - ❑This solution π is unique up to a multiplicative constant
  - ❑However, for every such solution, $\sum_i \pi_i = \infty$
- If it is transient, then:
  - ❑A non-zero solution to $\pi = \pi P$ may or may not exist, depending on the specific structure of the DTMC
  - ❑Even if a solution exists, the sum of its entries will again diverge, i.e., $\sum_i \pi_i = \infty$

Positive Recurrence vs. Null Recurrence of CTMC and Embedded DTMC

- Recall:
  - **Proposition**: If a semi-Markov process is irreducible and if T_{ii} has a non-lattice CDF with finite mean, then:
    - $P_i = \lim_{t\to\infty} P\{Z(t) = i | Z(0) = j\}$

    exists, is independent of initial state j, and:

    1) $P_i = \frac{\mu_i}{\mu_{ii}}$
  - If the embedded DTMC is irreducible and positive-recurrent, then:

    2) $P_j = \frac{\pi_j/v_j}{\sum_i \pi_i/v_i}$,
    - where the π_i are the stationary probabilities of the embedded DTMC, i.e., the unique non-negative solution of following system:
      - $\pi_j = \sum_i \pi_i P_{ij}$
      - $\sum_i \pi_i = 1$
- From 1) and 2), it follows that for an irreducible and positive-recurrent CTMC:

  3) $\frac{\mu_j}{\mu_{jj}} = \frac{\pi_j/v_j}{\sum_i \pi_i/v_i}$
- However, for the CTMC:

  4) $\mu_{jj} = E(S_{jj})$

  5) $\mu_j = \frac{1}{v_j}$
- By 3), 4), and 5), we get:

  6) $E(S_{jj}) = \frac{1}{\pi_j} \sum_i \frac{\pi_i}{v_i}$

Positive Recurrence vs. Null Recurrence of CTMC and Embedded DTMC (contd.)

- Recall: for an irreducible and positive-recurrent CTMC:

  6) $E(S_{jj}) = \frac{1}{\pi_j} \sum_i \frac{\pi_i}{v_i}$

- Equation 6) can also be shown (proof omitted) to be valid for irreducible and null recurrent CTMCs:
  - ❑ where $\pi > 0$ is a solution of the system of equations $\pi = \pi P$
  - ❑ in this case, LHS and RHS of 6) are ∞
- Hence, a recurrent CTMC is:

  7) positive recurrent if $\sum_i \frac{\pi_i}{v_i} < \infty$ and

  8) null recurrent if $\sum_i \frac{\pi_i}{v_i} = \infty$

- Recall that the embedded DTMC is positive recurrent iff $\sum_i \pi_i < \infty$, where $\pi > 0$ is a solution of the system of equations $\pi = \pi P$
- Hence, the condition for the embedded DTMC to be positive recurrent depends *only* on π
- However, from 7) and 8), it follows that the condition for positive recurrence of the CTMC depends on both π *and* $v_j, j \in S$
- Hence, in general, a recurrent CTMC and its embedded DTMC do not share the property of positive recurrence or null recurrence

Example

- Recall: for a birth and death process:

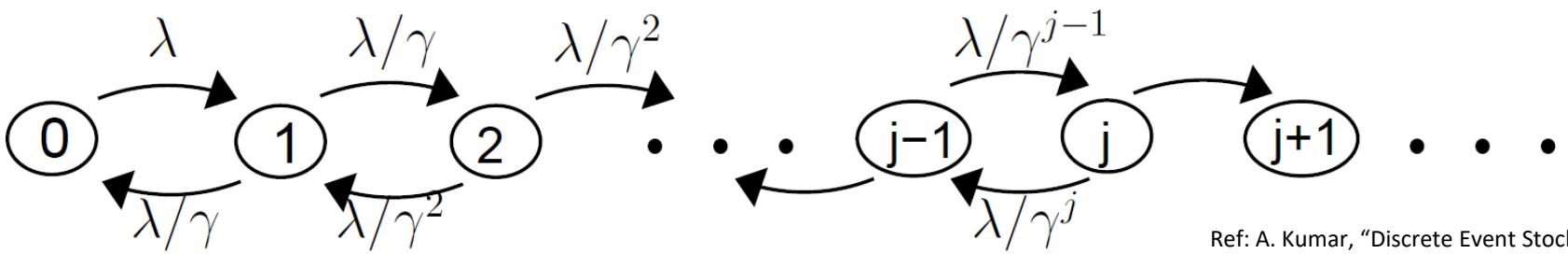
  1) $P_0 = \frac{1}{1+\sum_{n=1}^{\infty}\frac{\lambda_{n-1}\lambda_{n-2}\dots\lambda_0}{\mu_n\mu_{n-1}\dots\mu_1}}$

  2) $P_n = \frac{\lambda_{n-1}\lambda_{n-2}\dots\lambda_0}{\mu_n\mu_{n-1}\dots\mu_1\left(1+\sum_{n=1}^{\infty}\frac{\lambda_{n-1}\lambda_{n-2}\dots\lambda_0}{\mu_n\mu_{n-1}\dots\mu_1}\right)}$, for all $n \geq 1$

  - ❑ Condition needed for limiting probabilities to exist:

    3) $\sum_{n=1}^{\infty}\frac{\lambda_{n-1}\lambda_{n-2}\dots\lambda_0}{\mu_n\mu_{n-1}\dots\mu_1} < \infty$

- Suppose (see fig.):
  - ❑ $\lambda_0 = \lambda$
  - ❑ $\lambda_n = \mu_n$, for all $n \geq 1$
  - ❑ $\frac{\lambda_n}{\mu_{n+1}} = \gamma$, for all $n \geq 0$, where $0 < \gamma < 1$
- By 3), CTMC is positive recurrent
- Embedded DTMC:
  - ❑ has transition probabilities $P_{n,n+1} = P_{n,n-1} = \frac{1}{2}$ for all $n \geq 1$ and $P_{0,1} = 1$
  - ❑ is null recurrent
- Hence, here we have an example of a CTMC that is positive recurrent, but its embedded DTMC is null recurrent



Ref: A. Kumar, "Discrete Event Stochastic Processes"

Uniformization

- Consider a CTMC in which mean time spent in a state is same for all states, i.e.:
  1) $v_i = v$, for all i
- Let $N(t)$ be number of state transitions until time t
- $\{N(t), t \geq 0\}$ is:
  - a Poisson process with rate v
- Conditioning on $N(t)$, we get $P_{ij}(t)$:
  - $P\{X(t) = j | X(0) = i\}$
  - $\sum_{n=0}^{\infty} P\{X(t) = j | X(0) = i, N(t) = n\} P\{N(t) = n | X(0) = i\}$
  2) $\sum_{n=0}^{\infty} P_{ij}^{n} \frac{e^{-vt}(vt)^n}{n!}$,
  - where P_{ij}^{n} is the n-step transition probability of the embedded DTMC
- Equation 2) has limited applicability since it is based on the strong assumption 1)
- Next, we discuss how most CTMCs can be put in a form where 1) holds

Uniformization (contd.)

- Recall:

  2) $P_{ij}(t) = \sum_{n=0}^{\infty} P_{ij}^{n} \frac{e^{-vt}(vt)^n}{n!}$

- Consider a CTMC in which all the v_i are bounded by a positive constant, i.e.:

  3) $v_i \leq v$, for all i

- We introduce fictitious transitions at rate $v - v_i$ from each state i to itself
- So, transitions from every state i occur at rate v; out of these:
  - ❑ only a fraction $\frac{v_i}{v}$ are real transitions out of state i,
  - ❑ remaining fraction, $1 - \frac{v_i}{v}$, are fictitious transitions that leave process in state i
- Hence, the CTMC can be considered as a process that:
  - ❑ spends exponential amount of time with rate v in state i, and
  - ❑ then makes transition to j w.p. P_{ij}^{*}, where:

  4) $$P_{ij}^{*} = \begin{cases} 1 - \frac{v_i}{v}, & \text{if } j = i, \\ \frac{v_i}{v} P_{ij}, & \text{if } j \neq i. \end{cases}$$

- By 2), we have that the transition probabilities can be computed by:
  - ❑ $P_{ij}(t) = \sum_{n=0}^{\infty} P_{ij}^{*n} \frac{e^{-vt}(vt)^n}{n!}$
- This technique of uniformizing the rate at which a transition occurs from each state by introducing transitions from a state to itself is known as *uniformization*

Example

- Recall: we earlier considered a two-state CTMC, which:
  - spends an exponential time with rate λ in state 0 before going to state 1,
  - spends an exponential time with rate μ in state 1 before going to state 0
- Transition probabilities of embedded DTMC and transition rates:
  - $P_{01} = P_{10} = 1$, $v_0 = \lambda$, $v_1 = \mu$
- Let $v = \lambda + \mu$
- Uniformized version is a CTMC with:
  - $P_{00}^* = \frac{\mu}{\lambda+\mu} = 1 - P_{01}^*$,
  - $P_{10}^* = \frac{\mu}{\lambda+\mu} = 1 - P_{11}^*$,
  - $v_i = \lambda + \mu$, for $i \in \{0,1\}$
- Since $P_{00}^* = P_{10}^*$, it follows that probability of transition into state 0 is:
  - $\frac{\mu}{\lambda+\mu}$, regardless of present state
- Similar result holds for state 1
- Hence, n-step transition probabilities are given by:
  - $P_{i0}^{*n} = \frac{\mu}{\lambda+\mu}$, for $n \geq 1$ and $i \in \{0,1\}$
- $P_{00}(t)$:
  - $\sum_{n=0}^{\infty} P_{00}^{*n} \frac{e^{-(\lambda+\mu)t}[(\lambda+\mu)t]^n}{n!}$
  - $e^{-(\lambda+\mu)t} + \left[1 - e^{-(\lambda+\mu)t}\right] \frac{\mu}{\lambda+\mu}$
  - $\frac{\mu}{\lambda+\mu} + \frac{\lambda}{\lambda+\mu} e^{-(\lambda+\mu)t}$
- Similarly, $P_{11}(t)$:
  - $\sum_{n=0}^{\infty} P_{11}^{*n} \frac{e^{-(\lambda+\mu)t}[(\lambda+\mu)t]^n}{n!}$
  - $\frac{\lambda}{\lambda+\mu} + \frac{\mu}{\lambda+\mu} e^{-(\lambda+\mu)t}$